



## feature



### Astronauts unfurl ISS solar panels

On March 20, astronauts at the ISS unfurled the last set of panels, completing all the power generation modules on the space station

lunar outpost program will also take place over the same time period, with a permanent moon base established by 2025. The site for the NASA base has been identified as the Shackleton crater near the lunar south pole. The outpost is to be called the Neil Armstrong Lunar Outpost, and will function with a crew that has shifts more or less like the ISS. The RSA plans to send people to the moon by 2025, to establish a base that will begin construction well after the other nations. However, it will also be among the first to reach for Mars, along with JAXA and CNSA, by 2035.

The major factor delaying these plans is not a lack of technology, but a lack of funding. Because of the immense costs involved, independent approaches will not work out too well. The co-operation between ESA and RSA is pretty close, to the extent of sharing a launch pad. The RSA freely shares technology with the CNSA and ISRO. JAXA and ESA have a close association too. It makes economic sense for all these organisations to co-operate their efforts for the proposed base on the moon. Whether or not this is realized is yet to be seen. The alternative is another space race to establish the first base on the moon, which would be a costly and therefore slow affair for everyone involved. However, if the international co-operation on the ISS is any indication, the lunar base (or bases) will witness the same kind of international support.

#### 2030-2500

By this time, space agencies around the world will be thinking of serious colonisation of Mars. Once the Moon base has been established, the same technologies will be developed further for use on Mars. Due to the availability of water on the surface, a base close to the polar regions is very likely. Therefore, a colony on Mars is actually in one sense easier to set up, or at least maintain, than one on the Moon. A fleet of vehicles, possibly from different space agencies, will be used for the colonisation. Apart from research, mining is bound to be one of the primary operations on the Mars base. For this reason, some kind of capability of returning goods to Earth will have to be set up on Mars. This might include a space elevator, which involves having a cord running from the surface of the planet to a launch point in geosynchronous orbit.

Initially, enclosed habitats will be the only option. The bases on the moon and Mars are bound to grow in size over the next few decades. In effect, by the year 2080, at least one independent and permanent base will be established on Mars. The moons of Mars, Jupiter and Saturn are also candidates for temporary or permanent colonies in outer space. One of the most intriguing heavenly bodies in the solar system is Europa. This moon of

### Private spaceflight



In 1996, the X Prize foundation announced a US \$10,000,000 prize for the first private spaceship to reach space twice, in the span of a week – essentially for a re-usable craft. Later on, the prize was renamed Ansari X Prize. A company called Scaled Composites sponsored the Space Flight One project, which involved building a re-usable spacecraft like the Shuttle, but with many differences in the design. In June 2004, Astronaut Mike Melvill made the first sub-orbital space flight aboard the Space Ship One. He repeated the feat within the week to claim the Ansari X prize, and heralding the entry of the private sector into the arena of manned space flights.

The Virgin Group, collaborated with Scaled Composites and announced Virgin Galactic, the first commercial private space tourism enterprise. Previously, Space Tourists were taken aboard the ISS through the agency of Space Adventures, a subsidiary of the RSA. Virgin Galactic will build Space Ship Two, with more passenger capacity than Space Ship One. The Space Ship Two will be the first spacecraft built specifically for the purpose of tourism, and will take paid passengers to sub-orbital flights. Passengers will experience zero gravity for a very short time. The first flights are scheduled to be post 2010. The ticket rates initially are expected to be \$200,000, but will be reduced to \$20,000 after the first few flights. Eventually, Virgin Galactic is also planning to send spacecraft into orbital flights.

Projects under development are varied and many. The Romanian ARCAspace project has designs of a manned spacecraft by the name of STABILIO. The Blue Origin company, based in the US, has also tested a spacecraft called Goddard, their purpose being to launch sustainable sub-orbital spaceflights. XCOR, also a US based company, has various plans and designs for spacecraft.

Over the long term, many more private players are expected to get into the field, once a steady income base is established. Some private foundations are researching long term goals, such as a space ark to rescue humanity in case of a global calamity. The Lifeboat foundation has elaborate plans for this space colony, and can be viewed at <http://lifeboat.com>.

Jupiter is covered with ice, and many experts believe that under-water life may exist on this planet. If any method of generating fuel from water is developed, Europa is going to be the pit stop for access into the rest of the solar system and beyond. A colony on Europa will most certainly follow any successful activity on Mars. If mining takes off on a large scale on Mars, the asteroid belt will be a viable option for those companies or agencies that want to extract ores from space.

Meanwhile, a large-scale permanent space habitat in Earth orbit will be an option that will be explored by many private corporations. Apart from the establishment of space tourism with hotels in orbit, at least a part of solar energy harvesting and farming will take place in space. In a confined space, without seasonal changes, and 24x7 access to sunlight, crops could grow much better on farms in space. Solar panels in Earth orbit that beam back energy to the Earth in the form of microwaves might also be an energy efficient operation. The technologies for these, and the ideas, have been around since the 1980s.

Another interesting possibility will be independent and permanent human habitats at the various Lagrange points near the planet. A Lagrange point is a position between two heavenly bodies that can be occupied by a smaller body such as a space station. The point is such that it allows the smaller body to remain stable between the two heavenly bodies. Such points between the Sun and the Earth, and between the Moon and the Earth are candidates for permanent space habitats.

There are a number of proposed mechanisms for these permanent habitats. The ones being aggressively researched involve large torus structures that rotate at high speeds. A

Bernal Sphere or a Stanford Torus are large rotating structures that simulate gravity along the internal surfaces. Other design plans include beads, or spheres that are assembled in larger rings, or other configurations in outer space. This would allow easy fabrication from multiple locations around the world. Apart from gravity, these habitats would have to solve a number of problems, including recycling, atmospheric control and garbage disposal.

Cities in space, with anywhere between 10,000 to 1,00,000 people will be built at this point in time. This is an inevitable step according to experts, as such technology alone will ensure mankind's continuing existence beyond our home planet. The construction of these cities will be achieved by assembling parts in orbit. By this time, a space elevator from the surface of the Earth would considerably speed up this process. An interesting idea for a space elevator on the Earth involves a mobile sea tether